

1

CHARGING IS THE MOST DANGEROUS OPERATION IN THE LIFE OF THIS MACHINE — OUTDOORS, TRAINED OPERATOR, NO IGNITION SOURCE

CHARGING SEQUENCE

1

WELD OUT

Complete every wetted joint. No threaded or elastomer joint remains.

2

PROOF PRESSURE

Hydrostatic 21 bar, 5 min. Drain, dry with warm dry nitrogen.

1.5 x stagnation

3

LEAK TEST

Helium mass spectrometer under vacuum on the whole envelope.

< 1e-8 Pa.m3/s

4

BAKE-OUT

200 C / 8 h at < 1 Pa through the charge tube, pumping throughout.

carbon outgasses

5

COOL UNDER VACUUM

Let the bed reach ambient still under the pump. Do not vent.

no air, ever

6

CHARGE

Admit 4.9 kg anhydrous methanol by WEIGHT from a sealed burette.

by mass, not volume

7

DEGAS

Freeze-pump-thaw x3 on the receiver. Final pump to < 50 Pa at 0 C.

removes dissolved air

8

PINCH-OFF

Cold-weld the charge tube, cut, then TIG cap the stub.

permanent

9

LEAK TEST 2

Helium bomb 3 bar / 4 h, then mass spectrometer.

< 1e-8 Pa.m3/s

10

FIRST CYCLE

One full solar cycle under observation. Record bed and receiver temperatures.

commissioning

CHARGING RIG

Schematic

METHANOL BURETTE

COLD TRAP LN2 / dry ice

VACUUM PUMP 2-stage rotary

SI-100 MACHINE

The cold trap protects the pump and, more importantly, stops methanol vapour reaching the exhaust. Vent the pump outdoors, downwind.

CHARGING PARAMETERS		
PARAMETER	VALUE	LIMIT
Methanol charge	4.91 kg	by weight, ±1 %
Purity	anhydrous > 99.8 %	water poisons the carbon
Bake-out	200 C / 8 h	< 1 Pa
Final vacuum before seal	< 50 Pa	at 0 C
Proof pressure	21 bar	1.5 x stagnation
Leak rate	< 1e-8 Pa.m3/s	helium, twice
Seal	cold-weld + TIG cap	no valve, no port

PROHIBITED

Venting the envelope to air at any point after bake-out. Air in the bed is non-condensable gas and it never comes out again.

Charging by volume. Methanol expands 1.2 % per 10 K; the charge is a mass specification.

Any valve, gauge port or sight glass left on the wetted envelope. Every one is a leak path for the life of the machine.

Charging indoors, or anywhere without free ventilation and no ignition source within 5 m.

INDIA-VC THERMAL HARDWARE

ISO 128 / ISO 2768-mK

TITLE

VACUUM AND CHARGING

4.9 kg METHANOL / ONE-TIME PERMANENT CHARGE / NO FIELD SERVICE

SCALE

NONE

THIRD ANGLE PROJECTION

UNITS

mm

DRAWING No.

SI-200

REV

A

SHEET

5 / 6

DRAWN

AI-ASSISTED DRAFT

CHECKED

— PENDING —

MATERIAL

SEE BOM

FINISH

SEE NOTES

DATE

2026-09-15